

Axisymmetric Stokes Flow in FEniCS

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1 Introduction

Here, we consider a classic problem in Stokes flow - infinite fluid flow past a sphere with circulating fluid inside. The sphere is at rest while external fluid streams past it in the -z direction at velocity U from infinity. The flow is axisymmetric meaning it flows parallel to the axis of symmetry (z-axis) across all sides of the sphere. This also means there is no flow in the azimuth direction and velocity is independent of this angle. For velocity u and azimuth angle ϕ ,

$$u_\phi = 0 \quad \frac{\partial u}{\partial \phi} = 0 \quad (1)$$

which reduces the problem from three dimensions to a simpler two dimensions. With this two-dimensional problem setup, using the Stokes stream function to solve the problem is most efficient. Tangential velocity is continuous across its surface.

2 Governing equations

Starting from the equation

$$E^4 \psi = 0 \quad (2)$$

where

$$E^2 = \frac{\partial}{\partial \rho^2} + \frac{\sin \theta}{\rho^2} \frac{\partial}{\partial \theta} \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \right) \quad (3)$$

in spherical coordinates. E^2 can be thought of as similar to ∇^2 .

We guess that the general solution will be in the form

$$\psi = F(\rho) \sin^2 \theta \quad (4)$$

Now, we can find $F(\rho)$ by satisfying eq. 2.

$$E^2 \psi = \sin^2 \theta \left(F'' - \frac{2}{\rho^2} F \right) = f(\rho) \sin^2 \theta \quad (5)$$

The term $\left(F'' - \frac{2}{\rho^2} F \right)$ is represented as $f(r)$ because it simplifies the process as the equation is multiplied again by E^2 .

$$E^4 \psi = \sin^2 \theta \left(f'' - \frac{2}{\rho^2} f \right) = 0 \quad (6)$$

$$f'' - \frac{2}{\rho^2} f = 0 \quad (7)$$

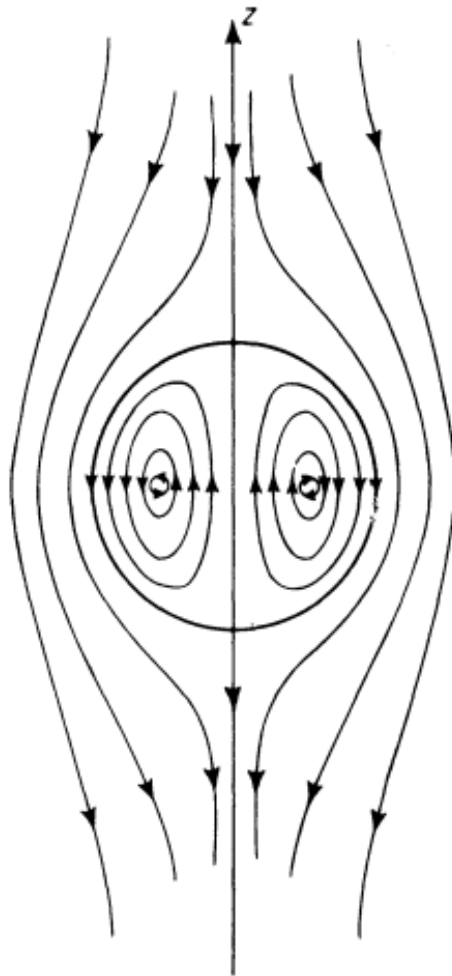


Figure 1: Fluid sphere with streamlines (Happel and Brenner)

By solving the ODE above, we obtain the general solution:

$$f(r) = A\rho^2 + \frac{B}{\rho} \quad (8)$$

Now substitute the $f(\rho)$ term to get

$$F'' - \frac{2}{\rho^2}F = A\rho^2 + \frac{B}{\rho} \quad (9)$$

Once again, solve the ODE to obtain the full general solution for $F(r)$

$$F(r) = \frac{1}{10}A\rho^4 - \frac{1}{2}B\rho + C\rho^2 + \frac{D}{\rho} \quad (10)$$

The general solution for the external and internal flow in spherical coordinates is

$$\psi^{(o)} = \sin^2\theta \left(\frac{1}{10}A\rho^4 - \frac{1}{2}B\rho + C\rho^2 + \frac{D}{\rho} \right), \quad \rho \geq a \quad (11)$$

$$\psi^{(i)} = \sin^2\theta \left(\frac{1}{10}E\rho^4 - \frac{1}{2}F\rho + G\rho^2 + \frac{H}{\rho} \right), \quad \rho \leq a \quad (12)$$

where a is the radius of the sphere, ρ represents the radial direction, and θ is the altitude angle. Expressions for velocity in spherical coordinates are

$$u_\rho = -\frac{1}{\rho^2 \sin(\theta)} \frac{\partial \psi}{\partial \theta} \quad (13)$$

$$u_\theta = \frac{1}{\rho \sin(\theta)} \frac{\partial \psi}{\partial \rho} \quad (14)$$

With these equations and the boundary conditions that follow, we can solve for the velocity vector field of the body of fluid. The first boundary condition for this problem is

$$\psi^{(o)} \rightarrow \frac{1}{2}U\rho^2 \sin^2\theta, \quad \rho \rightarrow \infty \quad (15)$$

which represents that the sphere is at rest while fluid streams past it with constant velocity U in the negative z direction. This leads to the values for the following coefficients:

$$A = 0 \quad C = \frac{1}{2}U \quad (16)$$

This is because ψ_o must match the first boundary condition as $r \rightarrow \infty$. Coefficients B and D don't affect this boundary because their terms approach 0 in the velocity expressions. Also, since velocity must be finite at the center of the sphere ($\rho = 0$), it becomes apparent that

$$F = 0 \quad H = 0 \quad (17)$$

The next boundary condition is

$$\psi^{(i)} = 0, \quad \rho = a \quad (18)$$

$$\psi^{(o)} = 0, \quad \rho = a \quad (19)$$

which corresponds to no penetration at the surface of the sphere.

$$\frac{\partial \psi^{(i)}}{\partial \rho} = \frac{\partial \psi^{(o)}}{\partial \rho}, \quad \rho = a \quad (20)$$

which assumes tangential velocity is continuous across the surface of the sphere.

$$\mu_i \frac{\partial}{\partial \rho} \left(\frac{1}{\rho^2} \frac{\partial \psi^{(i)}}{\partial \rho} \right) = \mu_o \frac{\partial}{\partial \rho} \left(\frac{1}{\rho^2} \frac{\partial \psi^{(o)}}{\partial \rho} \right), \quad \rho = a \quad (21)$$

which represents continuity of tangential stress at the boundary of the sphere.

Through systems of the general solutions and boundary conditions, the following coefficients are derived:

$$B = \frac{3}{2} U a \frac{1 + \frac{2}{3}\sigma}{1 + \sigma} \quad D = \frac{1}{4} U a^3 \frac{1}{1 + \sigma} \quad (22)$$

$$E = \frac{5}{2} \frac{U}{a^2} \frac{\sigma}{1 + \sigma} \quad G = -\frac{1}{4} U \frac{\sigma}{1 + \sigma} \quad (23)$$

where σ is the viscosity ratio of outer viscosity μ_o to inner viscosity μ_i .

$$\sigma = \frac{\mu_o}{\mu_i} \quad (24)$$

2.1 Transforming Coordinates

To solve this problem in the easiest way numerically, cylindrical coordinates should be used. Since there is no flow in the azimuth direction, cylindrical coordinates can be treated like two-dimensional Cartesian coordinates and are much simpler to write to FEniCS. The following equations are used to convert spherical coordinates to cylindrical coordinates:

$$\rho = \sqrt{r^2 + z^2} \quad (25)$$

$$\theta = \tan^{-1} \left(\frac{r}{z} \right) \quad (26)$$

3 Circulation inside a fluid droplet

The stream function governing flow inside a fluid droplet in the presence of bulk flow U from the surrounding fluid is

$$\psi^{(i)} = U r^2 \left(\frac{\sigma}{4a^2(1 + \sigma)} (r^2 + z^2) - \frac{\sigma}{4(1 + \sigma)} \right), \quad \rho \leq a \quad (27)$$

where r and z are cylindrical coordinates, σ is the ratio between the fluid viscosity outside and the fluid viscosity inside the droplet, and a is the radius of the droplet.

Accordingly, we can derive velocities in the r and z directions:

$$u_r = \frac{1}{r} \frac{\partial \psi^{(i)}}{\partial z} = \frac{U \sigma}{2a^2(1 + \sigma)} r z \quad (28)$$

$$u_z = -\frac{1}{r} \frac{\partial \psi^{(i)}}{\partial r} = \frac{U \sigma}{2(1 + \sigma)} \left(1 - \frac{2r^2 + z^2}{a^2} \right) \quad (29)$$

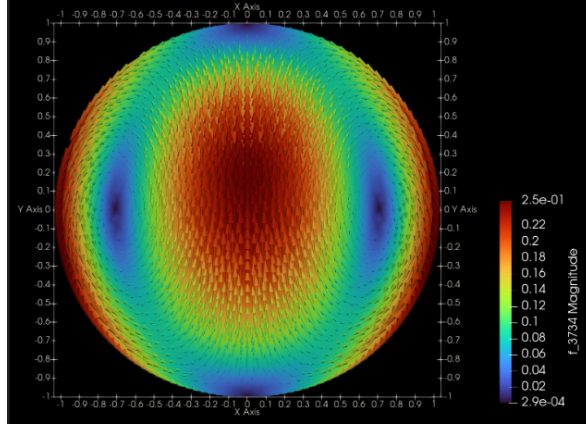


Figure 2: Fluid Circulation Inside the Sphere

4 Flow outside of the fluid droplet

The stream function for the region outside of the fluid sphere is defined as

$$\psi^{(o)} = Ur^2 \left(\frac{1}{2} + \frac{a^3}{4(1+\sigma)(r^2+z^2)^{\frac{3}{2}}} - \frac{3a(1+\frac{2}{3}\sigma)}{4(1+\sigma)\sqrt{r^2+z^2}} \right), \quad \rho \geq a \quad (30)$$

Once again, everything is converted to cylindrical coordinates. The velocity components are

$$u_r = \frac{3rz}{4(1+\sigma)} \frac{Ua}{(r^2+z^2)^{\frac{3}{2}}} \left(\frac{1+\frac{2}{3}\sigma}{(r^2+z^2)^{\frac{3}{2}}} - \frac{a^2}{(r^2+z^2)^{\frac{5}{2}}} \right) \quad (31)$$

$$u_z = \frac{Ua}{4(1+\sigma)(r^2+z^2)^{\frac{3}{2}}} \left(3r^2 + 2\sigma r^2 + 6z^2 + 4\sigma z^2 - 2a^2 + \frac{3a^2 r^2}{r^2+z^2} \right) - U \quad (32)$$

Here is how the solution of the entire system looks:

5 Implementing the Finite Element Method With FEniCS

5.1 Defining Gradient and Divergence

In the axisymmetric case, the gradient of a vector field A is given by the rank 2 tensor:

$$\nabla A = \begin{bmatrix} \frac{\partial A_r}{\partial r} & 0 & \frac{\partial A_r}{\partial z} \\ 0 & \frac{A_r}{r} & 0 \\ \frac{\partial A_z}{\partial r} & 0 & \frac{\partial A_z}{\partial z} \end{bmatrix} \quad (33)$$

In the same case of axisymmetry, the divergence of a vector A in cylindrical coordinates is defined as:

$$\nabla \cdot A = \frac{1}{r} \frac{\partial(rA_r)}{\partial r} + \frac{\partial A_z}{\partial z} \quad (34)$$

which simplifies to

$$\nabla \cdot A = \frac{\partial A_r}{\partial r} + \frac{A_r}{r} + \frac{\partial A_z}{\partial z} \quad (35)$$

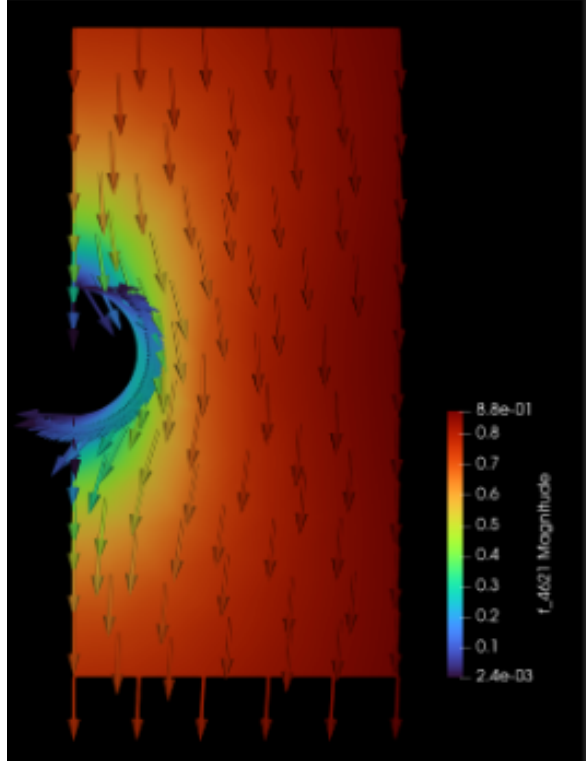


Figure 3: Flow outside of the sphere

via product rule.

5.2 Weak Form

For velocity, there are trial functions u and test functions v . For pressure, there are trial functions p and test functions q . The weak form is given by

$$a = \int_{\Omega} r \langle \nabla u, \nabla v \rangle d\Omega + \int_{\Omega} r p \nabla \cdot v d\Omega + \int_{\Omega} r q \nabla \cdot u d\Omega \quad (36)$$

$$L = \int_{\Omega} r \langle f, v \rangle d\Omega \quad (37)$$

where $a = L$.

Since we are dealing with no body forces,

$$f = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad (38)$$

5.3 Visualizing the Mesh (Figure got deleted)

To begin solving the finite element problem using FEniCS, we first created a mesh through Gmsh. Due to axisymmetry, the geometry of the problem originally a square containing a circle can be sliced in half and viewed as a semicircle within a rectangle.

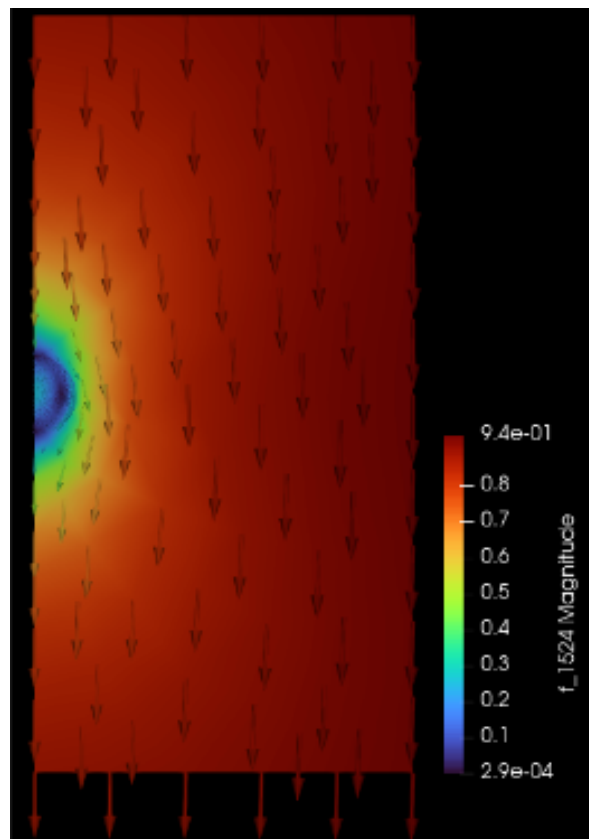


Figure 4: Full Analytical Solution

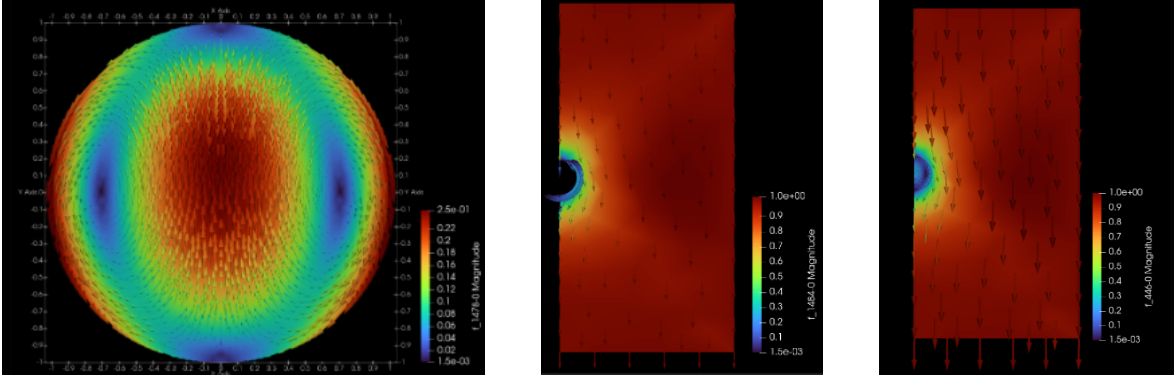


Figure 5: FEM solutions with prescribed boundaries

5.4 Boundary Conditions

The boundary conditions are defined a bit differently for the finite element problem. Along the top, right, and bottom boundaries, we set the condition $u_r = U$ where U is the velocity of the flow from ∞ in the negative z direction. Since in the analytical solution, the streamlines straighten out rather quickly and u gets very close to U , this boundary approximation is valid. Initially, the problem was solved by prescribing the analytical solution along the surface of the sphere.

$$u_r = \frac{1}{r} \frac{\partial \psi^{(i)}}{\partial z} = \frac{U\sigma}{2a^2(1+\sigma)}, \quad \rho = a \quad (39)$$

$$u_z = -\frac{1}{r} \frac{\partial \psi^{(i)}}{\partial r} = \frac{U\sigma}{2(1+\sigma)} \left(1 - \frac{2r^2 + z^2}{a^2} \right), \quad \rho = a \quad (40)$$

We use the velocity equations derived from the general solution of the sphere's inside, but it can work either way since the inside and outside equations are equal at the boundary. There is also no boundary condition needed for the axis of symmetry since it is automatically implied in the problem. Here are a few of the velocity vector fields that FEniCS outputs

Now, how can we solve this problem without knowing the analytical solution ahead of time? We need a boundary condition to represent the continuous behavior of velocity along the surface of the sphere.

$$u_\rho = 0, \quad \rho = a \quad (41)$$

$$\frac{ru_r + zu_z}{\sqrt{r^2 + z^2}} = 0, \quad \rho = a \quad (42)$$

represents no normal velocity. No fluid is entering or exiting the sphere.

$$u_\theta = a \frac{\partial v_\theta}{\partial \rho}, \quad \rho = a \quad (43)$$

$$u_r z - u_z r - \frac{ar z \left(\frac{\partial u_r}{\partial r} - \frac{\partial u_z}{\partial z} \right) - r^2 \frac{\partial u_z}{\partial r} + z^2 \frac{\partial u_r}{\partial z}}{\sqrt{r^2 + z^2}} = 0, \quad \rho = a \quad (44)$$

represents the tangential velocity along the surface. The tangential stress is continuous at all points of this boundary.

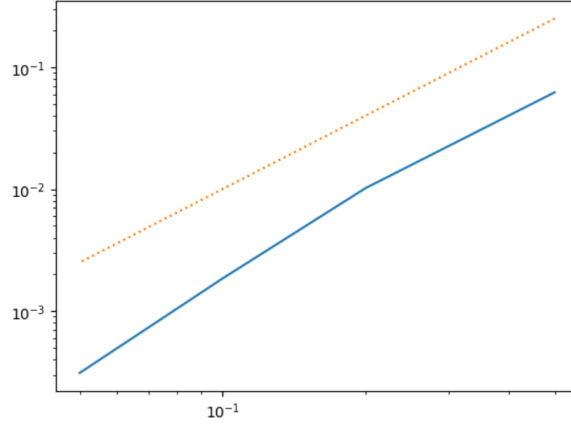


Figure 6: L2 vs Mesh Size

5.5 Results

We first considered the accuracy of our solution when prescribing its analytical form at the boundary of the sphere (See Figure 4). After prescribing analytical solutions at the boundary, we considered the accuracy of the solution within the domain by computing the L2 norm of the error:

$$L2 = \sqrt{\int_{\Omega} (u_r - u_{rFEM})^2 + (u_z - u_{zFEM})^2 d\Omega} \quad (45)$$

where u_r and u_z are analytical velocity components, and u_{rFEM} and u_{zFEM} are finite element velocity components. We plot L2 values over different time steps on a log-log scale.

The dotted line represents a slope of 2 and we can see that our L2 vs Mesh Size slope is around 2.

Finally, we considered a proof of principle for solving the full problem using finite element analysis. We first solved for flow in the outer domain by prescribing boundary velocity on edges not belonging to the symmetry axis or surface of the sphere. Over the spherical surface, we enforced zero normal velocity as specified by Eq 36. In particular, we added a term to our variational form:

$$R = \int_{\Omega} r \langle \nabla u, \nabla v \rangle d\Omega + \int_{\Omega} r p \nabla \cdot v d\Omega + \int_{\Omega} r q \nabla \cdot u d\Omega - \alpha \int_{\Gamma_{sphere}} ((ru_r + zu_z)(rv_r + zv_z)) r d\Gamma \quad (46)$$

where α is a constant used to penalize any deviations from zero normal velocity at the surface.

Using a value of $\alpha = 100$, we found a somewhat close solution, but there was slightly offset activity at the boundary of the surface.